

Class 10



Visiting the money metric indirect utility function to derive of the

welfare impact of a price Δ from P^0 to P' given the bench

mark price vector $\bar{P}$ $V_m(P', w | \bar{P}) - V_m(P^0, w | \bar{P}) = E(\bar{P}, V(P', w)) - E(\bar{P}, V(P^0, w))$

If positive, then the consumer is better off under P' than P^0 .

The sign is the same regardless of $\bar{P}$. but the size does depend on

$\bar{P}'$ simply because we are using different prices to evaluate

welfare impact.

2 bench marks are particularly natural $\bar{P} = P^0$ and $\bar{P} = P'$ i.e. evaluating

based on today's price.

If $\bar{P} = P^0$, the equivalent variation is

$$EV(P^0, P', w) = E(P^0, V(P', w)) - E(P^0, V(P^0, w))$$

$$= E(p^0, v') - E(p^0, v^0)$$

$$= E(p^0, v') - w$$

If under p^0 , the consumer receives $E(p^0, v') - w$. Then he can afford until v' because his

total wealth $E(p^0, v')$. So this is the amount of the consumer would

have to be paid today to be willing to forego

the price change. Hence it's the amount that

is equivalent to the price change.

If $\bar{p} = p'$, the compensating variation is $CV(p^0, p', w) = E(p', v(p', w)) - E(p', v(p^0, w))$

$$= E(p', v') - E(p', v^0)$$

If under p' today $w = E(p', v^0)$

$$= w - E(p', v^0)$$

you end up with wealth $E(p', v^0)$ which is just enough

to get you back to the old level of utility v^0 .

The amount to compensate for the change in prices.

Example $P_i' < P_i^0$ all else equal. The agent is better off $v' > v^0$.

Both EV and CV have to be positive. But likely to be diff.

Let's try to compare the size of EV and CV. From Shephard's Lemma, there's a

link between expend. func. and Hicksian demand. Let us use that link + link

between Hicksian and Marshallian demand in Slutsky equation.

$$\begin{aligned} \text{recall } EV(P^0, P', w) &= E(P^0, v') - w \\ &= E(P^0, v') - E(P', v') \end{aligned}$$

If only P_i changes using the fundamental of calculus

$$EV(P^0, P', w) = \int_{P_i'}^{P_i^0} \frac{\partial E(P, v')}{\partial P_i} dP_i = \int_{P_i'}^{P_i^0} H_i(P_1^0, P_2^0, \dots, P_{i-1}^0, P_i, v') dP_i$$

$\Rightarrow EV =$ are under hicksian demand curve evaluation

$$CV = \dots \dots \dots V^0$$

We need to compare hicksian demand at different V 's. We take a few steps.

1. Compare hicksian and marshallian demand for fix u . The Slutsky equation

$$\frac{\partial h_i(p, u)}{\partial p_i} = \frac{\partial x_i(p, w)}{\partial p_i} + \frac{\partial x_i(p, w)}{\partial w} x_i(p, w) \leq 0 \rightarrow \text{because concavity}$$

negative own price substitution effect - hicksian demand slopes.

At this point, add assumption that good i is a common good. $\frac{\partial x_i(p, u)}{\partial u} > 0$

$\Rightarrow \frac{\partial x_i(p, w)}{\partial p_i} < 0$ marshallian demand slopes down, it's ordinary good

$$\Rightarrow \frac{\partial h_i(P, v)}{\partial p_i} > \frac{\partial x_i(P, w)}{\partial p_i}$$

The left hand side is less negative than the RHS.

$\Rightarrow$ Hicksian demand is flatter.

$\Rightarrow$ The inverse hick demand curve is steeper.

2. An implication is that hicksian and marshallian demand. can cross only once.

What can we say about the price at which the crossing occurs and how it depends on utility

For concreteness, assume $p_i' < p_i^0$ and thus $v' > v^0$,

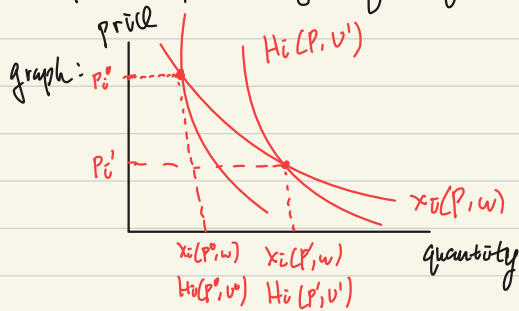
From Duality

$$H_i(p^0, v^0) = X_i(p^0, E(p^0, v^0)) = X_i(p^0, w)$$

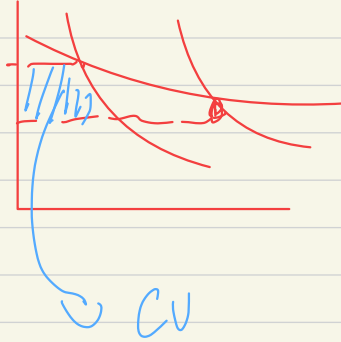
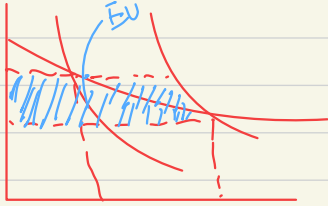
$$H_i(p', v') = X_i(p', E(p', v')) = X_i(p', w) > X_i(p^0, w)$$

Since good is an ordinary good.

Step 3: put everything together, compare EV and CV



Compare EV, CV Both = area under hicksian demand.



$EV > CV$.

EV and CV both measures change in indirect utility but in different ways sound concepts.

If you used area under Marshallian

demand - consumer surplus - to get either EV

or CV , conceptualization, what's the meaning?

CV is on shaky ground. The reason is that wealth

effects imply that Marshallian demand $\neq$ Hicksian demand.

The problem disappears if there're no wealth effects.

quasilinear preferences have this properties

- no wealth effect.

In this case $EV = CV = CS$ because three curves coincide.

$$V(x_i, x_j) = x_j + \hat{v}(x_i)$$

indiff. curve move in parallel way and a Δ in p_i

Does not impact consumption of x_i when intensify.